# Theodoros Mangas

## Backend Engineer — Python · Django · PostgreSQL

Komotini, Greece · Remote · open to EU/UK relocation · Employment or contract · theodoros-mangas.github.io/my-portfolio
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### PROFILE

Python/Django engineer who builds, ships and runs web applications end to end: data modelling, REST APIs, Celery background jobs, automated testing and Docker-based deployment on Linux. I have delivered two production client platforms and built an engineering SaaS on my own, from first commit to live operation. My background as a licensed land surveyor brings domain expertise in GIS, spatial data and engineering.

### EXPERIENCE

**Independent Python / Django Developer** — Freelance & own products — 2022 – Present · Remote

- Build and maintain Django applications end to end for clients and my own products: data model, backend, interfaces, deployment and ongoing support.
- Run production apps as Docker Compose stacks on Hetzner Cloud: Gunicorn behind a shared Nginx proxy, Let's Encrypt TLS, loopback-only ports, memory limits, and written deploy, backup and restore runbooks.
- **Spec-driven, AI-assisted workflow**: design docs and plans drive coding agents for implementation, tests and security review; I review every change, and CI checks each push.

### SELECTED SOFTWARE DELIVERY

**HandyTools** — Engineering & utilities platform (own product) — Jul 2025 – Present · restricted beta

Django 6 · DRF · Celery · Redis · PostgreSQL 17 · Docker Compose · Nginx · Leaflet

- Built solo (230+ commits): PDF, image and OCR tools plus a full-screen Leaflet **Surveyor Toolkit** for Greek cadastral work, in English and Greek, with Free/Pro/Enterprise tiers enforced server-side via DRF permissions.
- Designed **Celery/Redis** pipelines for OCR and PDF conversion with bounded retries for transient I/O errors, late acknowledgement, time limits and atomic result writes. Uploads up to 50 MB are staged on disk and passed to workers by path, keeping them out of Redis task messages. Fixed a retry bug that discarded successful results and refunded free-tier quota, locked in by a regression test.
- **Spec-first**: larger features start as a design doc and phased plan with explicit scope decisions. **Layered architecture**: framework-free, mypy-checked domain modules behind thin class-based DRF views; coordinate conversions route through one GGRS87 (ΕΓΣΑ87) pivot instead of pairwise transforms.
- **~700 Django tests** in GitHub Actions alongside ruff, mypy and a missing-migration check, with Docker images pushed to GHCR. Greek coordinate transformations (HEPOS, HATT) are tested to the millimetre against official values.
- Per-user quotas enforced under concurrency with **select_for_update** row locks; N+1 queries removed with select_related.

**Iasmos Culture Club Museum** — Paid client project — Delivered Sep 2026 · live · museum-iasmos-cultureclub.gr

Django 6 · SQLite · Docker · Gunicorn · Nginx · django-modeltranslation

- Bilingual (Greek/English) museum platform with a custom moderator CMS for exhibits, events, pages and media; soft delete with superuser-only restore.
- Privacy by design: self-hosted analytics with daily-rotating visitor hashes, consent-gated YouTube and Maps embeds, moderator Markdown sanitised with nh3, and django-axes login protection.
- **355 automated tests**; documented backup and restore procedures for the database and media.

**Agropixida** — Engineering practice platform — Live since 2022 · agropixida-syn2.gr

Django · DRF · MySQL · JavaScript · Leaflet · ezdxf · pyproj

- Bilingual business platform with a project showcase and a customer management area.
- Public engineering toolbox: DXF from coordinate files, and DXF/KML/KMZ/GeoJSON conversion with GGRS87 (ΕΓΣΑ87)/WGS84 reprojection.

**CityCare** — Volunteer team coordination app — Django 5 · Bootstrap · GitHub Actions

- Teams log actions at mapped locations; per-team daily statistics, rate-limited posts, Greek/English UI, 35 tests in CI.

### TECHNICAL SKILLS

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|---|---|
| **Backend & data** | Python, Django, Django REST Framework, Celery, Redis, PostgreSQL, MySQL, SQLite; ORM optimisation, row locking, rate limiting, django-axes |
| **Design & practice** | OOP (class-based views, abstract models, mixins, custom managers, dataclasses); layered architecture; spec-first design docs; AI coding agents (Claude Code) |
| **Delivery & quality** | Docker Compose, Linux, Nginx, Gunicorn, Git, GitHub Actions, Django test framework (unittest), ruff, mypy, Sentry |
| **Geo & frontend** | JavaScript, HTML/CSS, Bootstrap, Leaflet; pyproj, ezdxf, QGIS; DXF, KML/KMZ, GeoJSON, Shapefile |

### EARLIER & CONCURRENT EXPERIENCE

**Land Surveyor & Geoinformatics Engineer** — Independent practice, Greece — 2014 – Present · part-time

Surveying, CAD and GIS work in a multidisciplinary engineering team, plus automation for data processing and validation.

### EDUCATION, CERTIFICATION & LANGUAGES

**B.Sc. Surveying & Geoinformatics Engineering** — TEI of Central Macedonia, 2013

**Advanced Python Developer Certificate** — PeopleCert, 2022 · Greek (native) · English (fluent)